High performance computing

Introduction

Performance of the calculations

- Performance is (next to the accuracy, reliability, safety, ergonomics and ease of use and care) one of the most important characteristics of any software
- This lecture teaches how to analyze and achieve high performance in programs
- There is a specific domain of a computer science: High Performance Computing
- Results achieved in the HPC have their apply in all areas of computer science

HPC

- Achieving high performance
  - Application performance:
    - Languages and the environment with a high level of abstraction and complexity:
      - SQL and database systems
      - OpenGL, DirectX and graphics systems
      - HTML, PHP and web applications
      - Java, C #, and their execution environment
    - Languages relatively low level - close to the hardware
      - C
      - Assemblers
  - Due to the variety of languages and environments of high level and assemblers - far beyond time constraints, this lecture focuses on the C programming language

HPC

- Achieving high performance - C programs
  - Performance analysis for hardware architectures including processors (CPU, GPU), memory system, communication system
  - Optimization of instruction:
    - the source code - the assembler - execution by the processor
  - Optimizing memory access:
    - communication in systems with distributed memory
    - access to shared memory in multiprocessor and multicore systems
    - the functioning of the memory hierarchy in the CPU and GPU (cache memory, any other form of local memory, global memory)
    - hardware memory organization and its functioning

HPC

- High Performance Computing is a calculation in which we try to achieve the maximum processing speed
- Maximization of a processing speed has to lead to minimization of the duration of solution to the given problem (time-to-solution)
- Depending on the type of the calculation, we use the different measures of processing speed
- The most common measure (especially in the field of science computation) is the number of floating point operations performed per second (FLOPS)
- Other possible measures include the number of tasks in the second of instruction, the number of transactions processed per second, the number of displayed pixels per second, etc.

Performance of modern hardware

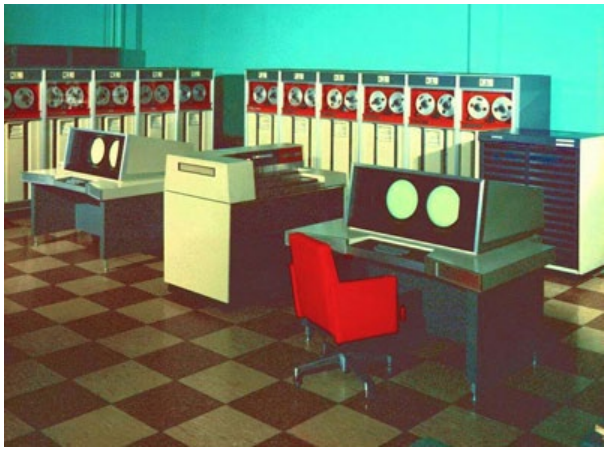
- CPU:
  - Intel i9-13900K – 1,11 Tflops
  - AMD Ryzen 9 7950X3D – 1,3 Tflops
- GPU
  - Geforce RTX 4090 – 82,58 Tflops (1,29 DP)
  - Geforce RTX 3060 – 12,74 Tflops (199 Gflops DP)
  - Geforce RTX 2080 Ti – 11.8 Tflops (367 Gflops DP)
  - Radeon RX 7900 XTX – 61,42 Tflops (1,919 DP)
  - Radeon RX 5700 XT – 9.8 Tflops (561.6 Gflops DP)
- Smartphones
  - iPhone 14 Pro Max – 1,789 Tflops
  - Samsung Galaxy S23 Ultra – 3,681 Tflops
- Console
  - PS4 – 1.84 Tflops (4.2 – Pro)
  - PS5 – 10.28 Tflops
  - PS5 Pro - 33.50 Tflops
  - Xbox One – 1.31 Tflops (1.4 - S)
  - Xbox Series X – 12 Tflops
  - Nintendo Switch – 393 Gflops (157 Gflops)
  - Nintendo Switch 2 – 3,1 Tflops (1,71 Tflops)

History of HPC

- matrix computers
- vector computers
- massively parallel computers
- clusters

1964 - CDC 6600

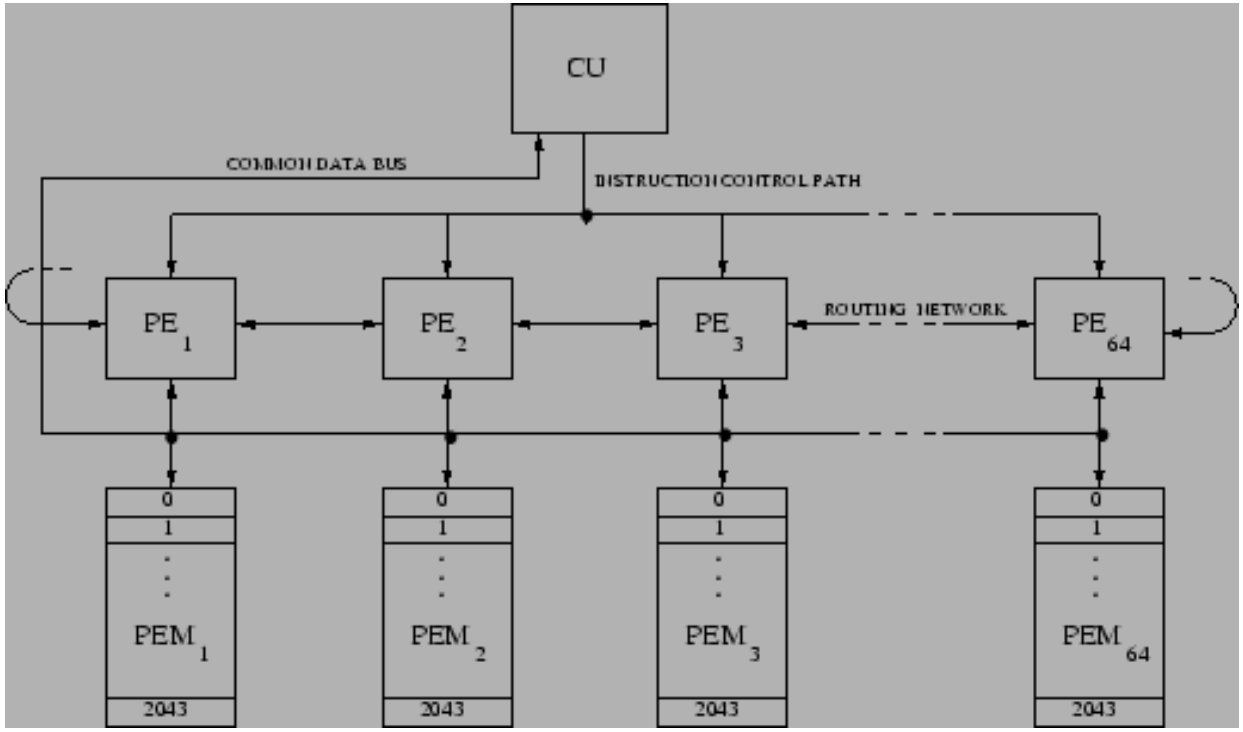
- **The First Supercomputer!**
- **1 CPU + 10 PP (Peripheral Processors)**
  - **CPU – Drawing – Space Sharing (RISC!)**
  - **PP – I/O Operations, Operating System (The Pioneer of Multithreading!)**



3 MFLOPS

Matrix computers (SIMD)

- Separate processor for processing orders
- Matrix of the processors for data processing



Illiac IV, 1966-76
~100 MFLOPS,
~5 000 000 USD

Matrix computers

- Today's applications:
  - SIMD orders in general purpose processors,
  - signal processing ,
  - GPUs,
  - special-purpose supercomputers (eg. apeNEXT - SPMD computer to simulation in the field of LQCD, 4096 processors, 7 TFLOPS)

Vector computers

- Basic element - vector processor:
  - vector registers
  - pipelining floating points operations
  - several streams and functional units
- Contemporary Applications:
  - superscalar processing and pipelining in general purpose processors
  - supercomputers

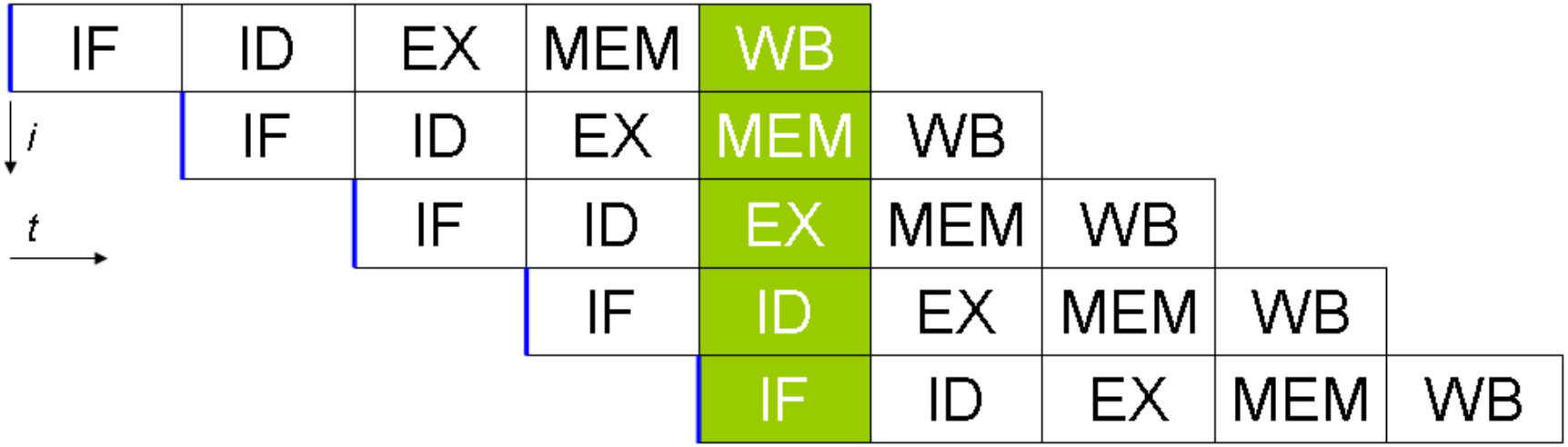
Cray-1, 1975
~100 MFLOPS,
~5 000 000 USD

Cray 2, ~2 GFLOPS

Vector computers

- scalar and vector units in processor (sometimes address)
- vector operations (including reduction operations)
  - op V -> V (e.g. a [i] := -b[i])
  - op V -> S - reduction (e.g. r := max (b[i]))
  - V op V -> V (e.g. a [i] := b [i] + c [i])
  - V op S -> V (e.g. a [i] := s * b [i])
- memory access: multislots, interlace, streams
- vector registers (length 64 to 128)
- pipelined functional units
- combining operations in chains (chaining) (a [i] := b[i] + s*c [i])

Pipeline


IF ID EX MEM WB
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Superscalar

IF	ID	EX	MEM	WB				
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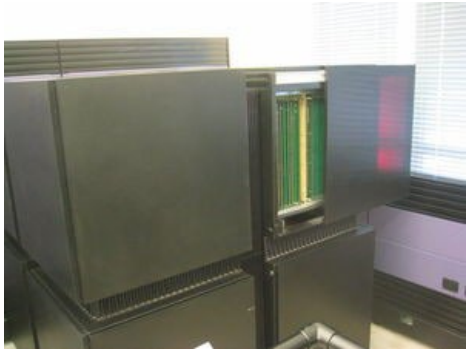
Massive-parallel computers

- Computers having hundreds, thousands or more processors (vector computers - tens)
- Computers composed of independent processors, most often equipped with individual memory

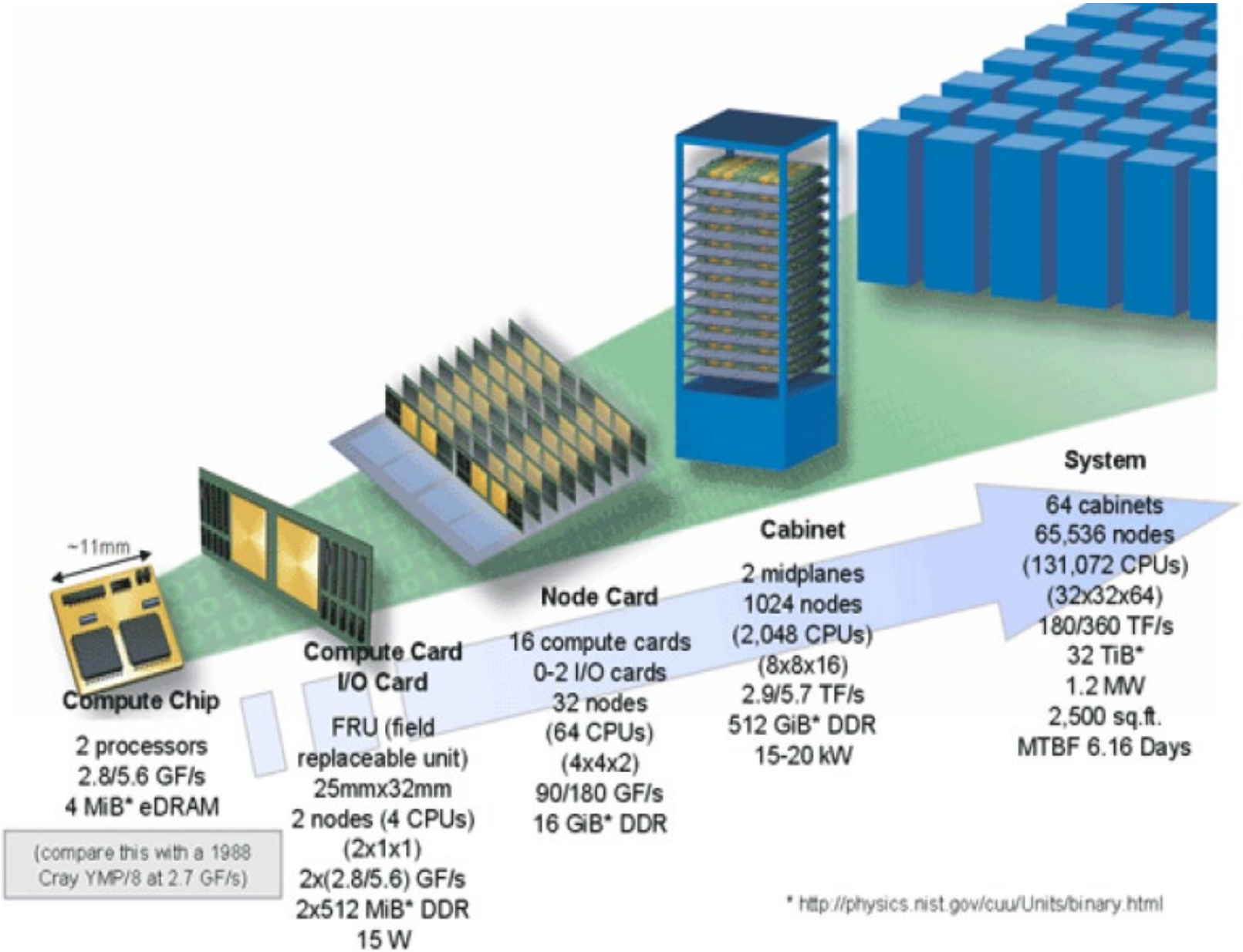


CM-2, 1987
65 536 processors

CM-5, 1991, 512 nodes
~65.5 GFLOPS,~25 000 000 USD

IBM Blue Gene/L

- *Blue Gene Architecture*
- *65 536 computing nodes*
- *280 TFLOPS*
- *64 racks*
- *1.2 MW*
- *2004 - 2007*


~11mm
Compute Chip
2 processors
2.8/5.6 GF/s
4 MiB* eDRAM
(compare this with a 1988 Cray YMP/8 at 2.7 GF/s)
Compute Card
I/O Card
FRU (field replaceable unit)
25mmx32mm
2 nodes (4 CPUs)
(2x1x1)
2x(2.8/5.6) GF/s
2x512 MiB* DDR
15 W
Node Card
16 compute cards
0-2 I/O cards
32 nodes
(64 CPUs)
(4x4x2)
90/180 GF/s
16 GiB* DDR
Cabinet
2 midplanes
1024 nodes
(2,048 CPUs)
(8x8x16)
2.9/5.7 TF/s
512 GiB* DDR
15-20 kW
System
64 cabinets
65,536 nodes
(131,072 CPUs)
(32x32x64)
180/360 TF/s
32 TiB*
1.2 MW
2,500 sq.ft.
MTBF 6.16 Days
* http://physics.nist.gov/cuu/Units/binary.html

Clusters

- independent computers connected by a network
- strong integration through special software
- for several thousand computers
- relatively inexpensive (a few thousand USD / PC)

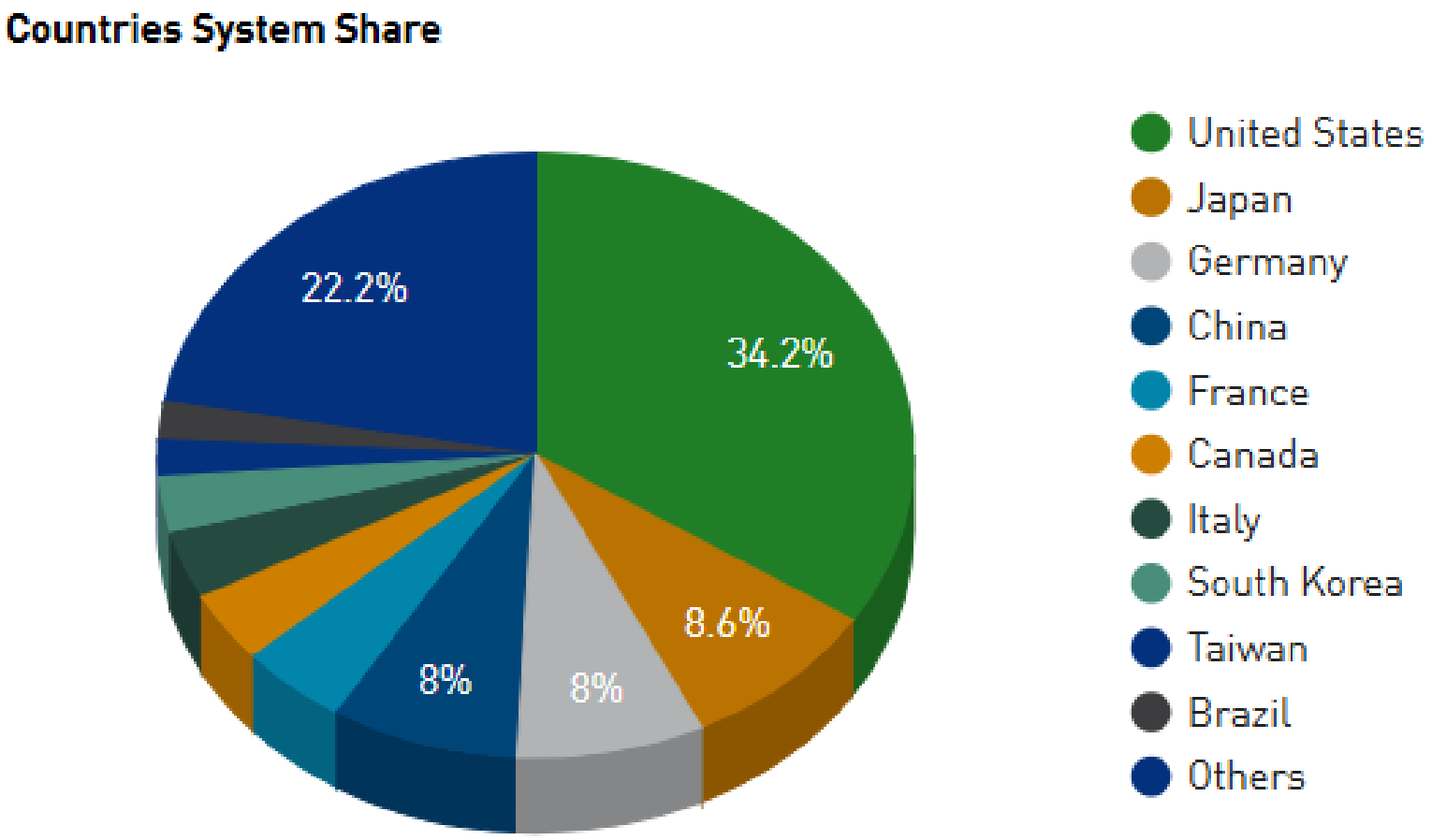
And the winner is...

- *TOP500 - the most popular classification the most powerful computer systems in the world*
- *Carried out twice a year (in June and November) since 1993*
- *The classification is based on the speed of solving the system of linear equations using Gauss method*
- *Only general-purpose systems*

Top 5

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	**JUPITER Booster** - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794
5	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	

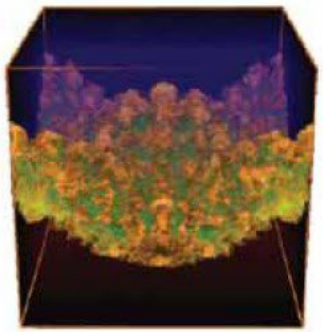
Countries


Countries System Share
22.2%
34.2%
8.6%
8%
8%
United States
Japan
Germany
China
France
Canada
Italy
South Korea
Taiwan
Brazil
Others

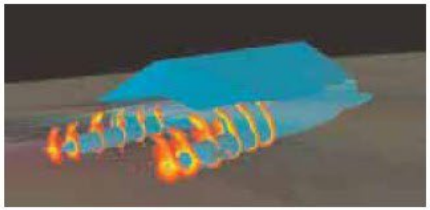
Is that all?

- Systems placed on the TOP 500 list are not the only powerful computing systems in the world
- Equally powerful tend to be organized ad hoc teams of computers distributed around the world and connected by a network
- Project SETI@home (now as BOINC, http://boinc.berkeley.edu/), Folding@home project (http://folding.stanford.edu/):
  - hundreds of thousands of computers around the world
  - several PFLOPS total computing power

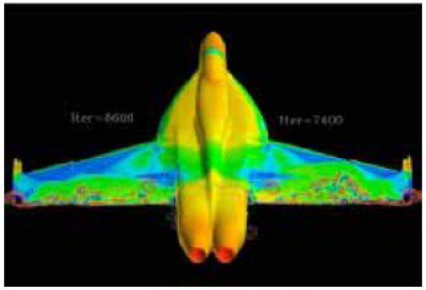
Where we need HPC



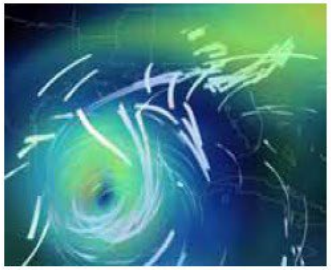
Nuclear Stockpile Stewardship



Ship Design

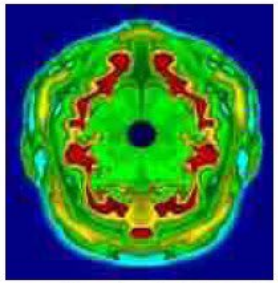


Aeronautics

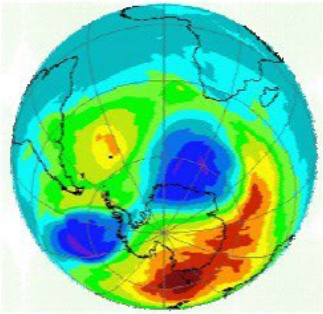


Weather Prediction

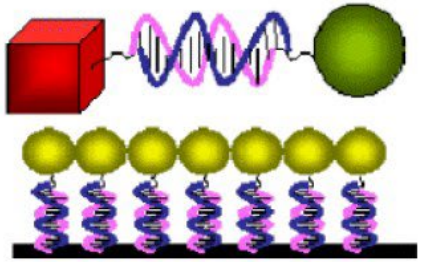
Cryptanalysis



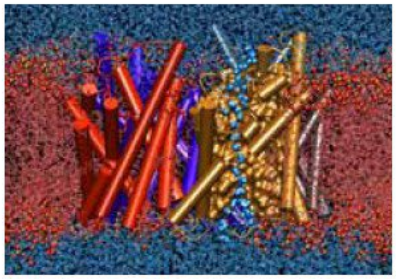
Astrophysical Simulation



Climate Modeling



Nano-Science



Biology